

Ishita Sinha

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Software Engineer with strong ML & AI skills | ~1.5 years of experience across full-stack development and applied ML.

Education

Cornell University | **M.Eng. Computer Science** Aug 2025 – May 2026
Machine Learning, Computer Vision, Systems for large-scale ML, Software Engineering, Deep Learning. GPA: 3.9/4.0

Sikkim Manipal University (SMU) | **B.Tech. Computer Science & Engineering** 2020 – 2024
University Gold medalist in all 4 years of B.Tech. GPA: 4.0/4.0

Skills

Programming: Python, Java, C, C++, C#, .NET, JavaScript, DSA (150+ LeetCode)
Core ML: Bayesian models, KNN, SVM, gradient descent, PCA, clustering, Reinforcement Learning, Hugging Face
DL: Neural networks, Transformers/LLMs, CUDA/Multi-GPU, vLLM, NCCL, NVSHMEM, LoRA, DoRA, PyTorch, TensorFlow
Computer Vision: 3D reconstruction, segmentation, CNN, CLIP, NeRF, ViTs, Diffusion Model, GANs, VLMs
AI Systems: AI agents, statistical modelling, LangChain, Vector DB, OpenAI APIs, Azure AI/Cognitive Search
Systems: OS, networks, DBMS, software engineering, compilers, computer architecture, Linux/Windows/Mac
Full Stack: Angular, React, Node.js, HTML, CSS, Spring Boot, Flask, Streamlit, MongoDB, MySQL
Cloud & DevOps: Git/GitHub, Azure, Apache Airflow, Google Cloud, CI/CD (YAML), HPC (Nvidia Perlmutter)

Work Experience

Microsoft | **Software Engineer** Aug 2024 – Apr 2025

- Developed a cloud-based automation testing framework for BDO Global's accounting platform, integrated into Azure DevOps workflows using .NET, C#, and Playwright, increasing test coverage; additionally built an open-source prototype using Java and Gradle.
- Built and deployed a microservices-based full-stack e-commerce system using Angular, Node.js, and Azure SQL on Azure Cloud, enabling automated CI/CD deployments and secure JWT-based authentication.

Keysight Technologies | **R&D Co-op** Jan 2024 – Jun 2024

- Built an application to verify Time-Sensitive Networking (TSN) packet exchanges over a distributed Wi-Fi network as per IEEE 802.1AS in Python, FastAPI, and PyShark. Analyzed fine-time measurement frames using Wireshark/tshark. Developed CI/CD pipelines using YAML. Contributed to its patent submission for Qbv-over-Wi-Fi.
- Enabled detection of traffic-flow delays in ML collective communication over RoCE v2, improving performance analysis of distributed workloads using Pandas, Docker and Pytest

Microsoft | **Software Engineer Intern** Jun 2023 – Jul 2023

- Built a conversational AI chatbot for cloud migration teams using Azure OpenAI, Azure Cognitive Search, LangChain, and ChromaDB for contextual search across manuals and repositories.
- Delivered a cloud deployed application Azure DevOps, Python, Flask, Streamlit and Azure Blob Storage.

Projects

Research Project - Numerical Stability for ML Systems (*Planned publication in ICML*) Sep 2025 – May 2026

- Improved numerical instability arising in kernel-based ML methods under clustered or nearly coincident data points.
- Significantly improved conditioning of Gaussian kernels using stabilized kernel differencing and preconditioning methods.

Open Source Project - Apache Airflow provider health dashboard Jan 2026 – May 2026

- Building an open-source dashboard to assess the health and maintenance of cloud and ML integration providers to Apache Airflow. Analyzes GitHub repository signals like issue resolution time, PR latency, and contributor activity.
- Enables production-readiness scoring and proactive risk detection to support data-driven governance.

Nvidia LLM Benchmarking Project - Distributed LLM communication Nov 2025 – Dec 2025

- Benchmarked GPU communication for distributed LLM training (Llama 3-8B) on Nvidia Perlmutter using Torch Distributed/TorchTitan, comparing NCCL, NCCL-GIN, and NVSHMEM across data, tensor, and pipeline parallelism
- Achieved 160% higher DP throughput and 53% better inter-node performance through communication–compute overlap.

Image-Based Cryptography & Deep Learning Jan 2023 – May 2023

- Built an image-based secret-sharing cryptography application using Python, OpenCV, and NumPy. Built CNN models using TensorFlow to improve encryption quality.